FIGURE 3: Hybrid Clinical Decision Workflow, Platt Calibration & Penetrance Mapping

Deterministic clinical safeguards, 5-seed neural ensemble inference, empirical calibration, and zygosity-aware risk assignment

Patient Genomic VCF
Input VCF (.vcf / .vcf.gz)
Strict GRCh38 Reference Check
Quality Filters: PASS
Read Depth DP ≥ 10, GQ ≥ 20
Sequencing Quality Control

Stage I: Population Frequency
Query gnomAD Global Cohort
Condition: AF > 0.01 (1.0%)?
• YES: Common Neutral Polymorphism
• NO: Advance to Clinical Curations
Unclassified SNVs
Deterministic Threshold

NO

Stage II: ClinVar Expert Consensus
Index: (chr, pos, ref, alt)
Evidence Stars ≥ 2 Stars?
(Multiple submitters / Expert panel)
• YES: Adopt Curated Consensus
• NO: Novel or 1-Star/Conflicting
Database Ground-Truth

NO

Stage III: Structural LoF & Splicing
Structural Rupture Rules:
1. Frameshift or Canonical Splice (±1,2 bp)
2. SpliceAI DS_MAX ≥ 0.50 (Cryptic)
• YES: Structural Inactivation Rule
• NO: Unclassified SNV
Rule-Based Inactivation

YES (AF > 1%)

Verdict: Benign
Common Polymorphism
Calibrated Score: $P = 0.001$
Zero Clinical Penetrance
Resolved: Benign

YES (Stars >= 2)

Adopt ClinVar Verdict
Pathogenic: $P = 0.999$
Benign: $P = 0.001$
VUS: $P = 0.500$
Curated Consensus

YES (LoF)

Verdict: Likely Pathogenic
Predicted Loss-of-Function
Calibrated Score: $P = 0.950$
Bypasses Protein Model
Resolved: LoF Rule

Stage IV: FusionResNet Ensemble (5 Seeds)
Multimodal Feature Fusion (Tabular + ESM-2)
Multi-Seed Averaging (Seeds 42, 101, 2024, 7, 999):
$z_{\text{ensemble}} = \frac{1}{K} \sum_{k=1}^{K} z^{(k)}$
Out-of-Gene AUC = 0.9919 [95% CI: 0.9913–0.9925]
Raw Logit: $z_{\text{ensemble}} \in \mathbb{R}$

Stage V: Empirical Platt Calibration
Optimized over Validation Logits via L-BFGS:
$z_{\text{cal}} = s \cdot z_{\text{ensemble}} + b$
($s = 1.0559$, equivalent $T = 0.947$, $b = -1.5561$)
Posterior Probability: $P = \sigma(z_{\text{cal}})$
Expected Calibration Error: ECE : $0.0279 \rightarrow 0.0009$
Calibrated Probability: $P \in [0, 1]$

Stage VI: ACMG/AMP Tier Assignment
• $P \geq 0.90 \rightarrow$ Pathogenic
• $0.70 \leq P < 0.90 \rightarrow$ Likely Pathogenic
• $0.30 \leq P < 0.70 \rightarrow$ Variant of Uncertain Significance (VUS)
• $0.10 \leq P < 0.30 \rightarrow$ Likely Benign
• $P < 0.10 \rightarrow$ Benign
Categorical Decision Boundaries

Stage VII: Zygosity Resolution & Curated Lifetime Penetrance Mapping (Hospital PDF Output)
1. Autosomal Dominant Genes (e.g., BRCA1, BRCA2, TP53, MLH1, MSH2): Monoallelic Pathogenic $\rightarrow$ Immediate High-Penetrance Assignment
2. Autosomal Recessive Genes (e.g., MUTYH, NTHL1): Monoallelic Heterozygous $\rightarrow$ Reclassified as 'Carrier (Recessive)', Biallelic/Compound Het Required for High Risk
3. Curated Penetrance Lookup (data/penetrance.csv): JAMA/Lancet/NEJM peer-reviewed lifetime cancer risk ranges vs. general population baseline
4. Output Deliverables: Structured JSON record, Plaintext clinical summary, Publication-grade Hospital PDF Report with Visual Risk Radar
Final Clinical Translation: Actionable Risk Stratification